

Role of forms in webpages:

- > HTML forms are used to take user inputs.
- > With the help of forms we can collect the data from all the users at once.

HTML Forms:

- > To provide forms in webpage we have to use `<form>` tag.
- > action $\Rightarrow$ It is used to provide server reference to which data has to be sent.

Form Elements:

- > To take different inputs, form provides different elements.
- > To take inputs from the user we have to use `<input>`.
- > `<input>` tag has attributes type, name and id.
- > type attribute decides which different elements to take different inputs.
- > name is used to provide name and id is used to provide unique name.
- > Form elements consist of:

1. `<input>` $\Rightarrow$ Used to create various types of user input fields (text, email, password, etc.)
2. `<label>` $\Rightarrow$ Defines a label for an input element to improve accessibility
3. `<button>` $\Rightarrow$ Creates a clickable button for actions like submit or reset
4. `<textarea>` $\Rightarrow$ Used to take multi-line text input from the user (address, feedback, etc)
5. `<fieldset>` $\Rightarrow$ Groups related form elements together inside a box (border)
6. `<legend>` $\Rightarrow$ Provides a caption/title for a `<fieldset>`
7. `<select>` $\Rightarrow$ Creates a dropdown list for selecting options
8. `<option>` $\Rightarrow$ Defines individual items inside a dropdown (`<select>`)
9. `<optgroup>` $\Rightarrow$ Groups related options inside a dropdown

10. <datalist> => Provides predefined suggestions for an <input> field

What is <select>?

-> A <select> tag is used to create a dropdown list where the user can only choose from predefined options.

-> Only select from list ✕ typing not allowed.

-> For Example:

```
<label>Select your country:</label>
<select>
  <option>India</option>
  <option>USA</option>
  <option>UK</option>
</select>
```

What is <datalist>?

-> A <datalist> tag is used to show suggestions in an input field while still allowing the user to type their own value.

-> Suggestions + typing allowed ✓

-> For Example:

```
<label>Choose your browser:</label>
<input list="browsers">
<datalist id="browsers">
  <option value="Chrome">Chrome</option>
  <option value="Firefox">Firefox</option>
  <option value="Edge">Edge</option>
</datalist>
```

Input Types:

1. `<input type="button">`
2. `<input type="checkbox">`
3. `<input type="color">`
4. `<input type="date">`
5. `<input type="date-time local">`
6. `<input type="email">`
7. `<input type="file">`
8. `<input type="hidden">`
9. `<input type="image">`
10. `<input type="month">`
11. `<input type="number">`
12. `<input type="password">`
13. `<input type="radio">`
14. `<input type="range">`
15. `<input type="reset">`
16. `<input type="search">`
17. `<input type="submit">`
18. `<input type="tel">`
19. `<input type="text">`
20. `<input type="time">`
21. `<input type="url">`
22. `<input type="week">`

Input Attributes:

1. type => Defines the type of input (text, email, password, number, etc.)
2. id => Unique identifier for an element (used for CSS/JS and linking with label)
3. name => Name used to identify form data when submitting
4. placeholder => Shows hint text inside input before user types
5. required => Makes the field mandatory to fill before submitting
6. value => Sets the default value of an input field
7. readonly => Makes input non-editable but still submits value
8. disabled => Disables the input (cannot edit or submit value)
9. size => Sets the visible width of input (in characters)
10. autofocus => Automatically focuses the input when page loads
11. min & max => Sets minimum and maximum values for input (mainly numbers/dates)
12. minlength & maxlength => Sets minimum and maximum number of characters allowed
13. multiple => Allows selecting multiple values (for file or select inputs)
14. list => Connects input with <datalist> for suggestions
15. step => Defines interval between allowed values (e.g., 1, 2, 0.5 etc.)